A Mathematical Modeling for the Infection and Recover Rates of COVID-19

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***Abstract*— The novel coronavirus, COVID-19, has swept nations since November, 2018, leaving countless communities devastated in its wake. Like many countries, the United States continue to deal with the outbreak, imposing policies and regulations such as "stay-at-home" and "social distancing" orders in an attempt to prevent urgent hospitalizations from the virus to reach unmanageable levels and overwhelm health care facilities. In this paper, we apply a logistic regression formula to case data from three different communities (Denver, CO; Jacksonville, FL; and New York, NY) and attempt to simulate this form using agent-based modeling in NetLogo. Using our mathematical formula, we have accurately described the total cases and total death rates for these three communities; our simulation showed less accuracy, which may be due to assumptions made when creating the model.**

Keywords: Coronavirus, Virus, Disease, Logistic Regression, Agent-Based Modeling

I. INTRODUCTION

This paper aims to predict total case and mortality rates of COVID-19 in three communities in the United States–Denver, CO; Jacksonville, FL; and New York, NY, using logistic regression and agent-based simulation. Per The Center for Systems Science and Engineering at Johns Hopkins, the United States currently surpasses all other nations in the number of confirmed cases, reaching 1,056,402 as of 30 April, 2020. Furthermore, the United States is experienced high mortality rates from the novel virus, with death counts in New York City, New York alone totaling 18,069 as of 30 April, 2020 [1]. In our research, we examined the various policies set in place by local governments in order mitigate spread of the disease and prevent shortages of healthcare resources. These policies include "stay-at-home" orders and "social-distancing" protocols.

The following chosen data sets are pertaining to Denver, Jacksonville, and New York City–the highest population locations in Colorado, Florida, and New York, respectively. The data sets used for this research have been organized around the themes of COVID-19 case and mortality rates within these locations, and the local policies set in place in order to "flatten the curve"[2], or mitigate high concentrations disease onset at one time. These data sets are critical when addressing the issue of case and mortality rates from the novel coronavirus because it provides real time data on testing, positive cases, and mortality from which a model may be created, as well as providing guidelines for population movement so that research on the spread of this virus may be more accurately predicted for each region. The rapid spread of the novel virus may be a result of high population density in conjunction with lack of social-distancing measures including social isolation and quarantine. Along with pertinent information regard current case and mortality rates in each selected location, it was imperative for us to gather data on the government-mandated social distancing practices of each region to explore how these regulations may impact disease spread and therefore, change how we chose to create our model. Furthermore, this information would help with data and model analysis, as we would be able to determine if these practices will prove effective in lowering overall infection and mortality rates, or have any impact on the speed in which the virus is spread.

Because of the nature of the problem at hand, this paper and the corresponding mathematical model and simulation is most pertinent to individuals within the regions studied; particularly healthcare professionals and government officials who must now make decisions on how to implement social distancing guidelines as the United States enters another month of dealing with the pandemic and virus spread. This audience would benefit most from this research because questions of how virus spread may be impacted by social-distancing measured, or lack thereof, may be answered through this model and the following analysis. Additionally, individuals who wish to better understand the implications of social-distancing measures and the predicted spread of the virus may gain insight from this model, as it provides invaluable data on the current state of cases and mortality in our three selected regions, and may provide fairly accurate predictions on the spread henceforth.

II. METHODS AND PROCEDURES

A. Data Sources

13 sources were evaluated, including as 2 data sets, originating from 1Point3Acres, World Health Organization, and The Center for Systems Science and Engineering at Johns Hopkins, among other credible and peer-reviewed sources. A thorough literature was conducted; however, for the sake of brevity, only pertinent data sources will be discussed here.

1Point3Acres provided credible real-time updates on coronavirus cases in the United States and Canada. Information provided included total and new infected cases, total and new recovered cases, and current and new deaths attributed to the virus. Additionally, the source provides real time visualization via graph of each of these features, both cumulative and new. Data for this resource is aggregated from official government websites and new reports[3].

Additionally, information was collected from The Center for Systems Science and Engineering at Johns Hopkins.

This source provided similar information to 1Point3Acres, including total deaths for several countries in addition to the United States, and a break down of total deaths and recoveries by city and state. This source also includes the total number of individuals tested in each state, along with the total number of hospitalizations per state. Like 1Point3Acres, this source provides visualizations of the spread with graphs and maps. Furthermore, this source gives qualitative information, such as analysis on where positive cases are increasing, how the outbreak is growing, and how mortality rates differ across countries [1].

Original users of these data sets include government officials reviewing the data to make policy decisions, healthcare workers looking to track the trends of the virus, and general public aiming to keep updated on the state of cases in their respective regions and nationally.

B. Conceptual Model

In our research, our aim was to track the path of the novel virus through 3 separate regions in the United States. To effectively do this, we chose to identify our state variables as follows (see fig. 1):

- Susceptible Population (S): Population susceptible to virus
- Infectious Population (I): Infected population able to transmit the disease to other individuals
- Recovered Population (R): Population that has recovered from the disease, have died from the disease, or are otherwise immune to the disease and thus cannot contribute to the transmission of the virus

Furthermore, we have identified the following parameters:

- Transmission (α)
- Recovery to original health of death (β)

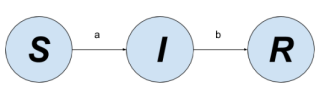

Fig. 1: Conceptual model, where S is the susceptible population, I is the infected population, R is the recovered population, α is infection, and β is recovery

For this research, several assumptions were made regarding the data and model [4].

1) Zoonotic transmission associated with COVID-19 is not possible. The virus may only effectively be transmitted from human to human, and animals play no role in spreading the virus.
2) The incubation period of COVID-19 is up to 14 days. [2] The World Health Organization estimates that the incubation period ranges from 1-14 days, with median estimates of 5-6 days between infection and the onset of clinical symptoms of the disease.
3) Any person who is in close contact (within 6 feet) of an individual with a COVID-19 case is at risk of being exposed to infectious respiratory droplets.
4) Cases are no longer communicable 10 days after onset of illness.
5) Individuals with a COVID-19 case will infect an estimated 3.87 other individuals (CI: 95%, 2.5-4.7).

After determining our model assumptions, we choose three different strategies for our conceptual model: (1) No mitigation strategy, (2) a stay-at-home order followed by no physical distancing, and (3) a stay-at-home order followed by physical distancing.

C. Mathematical Model

For our mathematical model, we chose to first evaluate data from 1Point3Acres concerning total cases and total deaths for New York, Florida, and Colorado. Data obtained was between the span of 18 March through 16 April (see figures 2,3, and 4 for data).

New York COVID–19 Total Cases and Total Deaths Data

Date:	Total Cases:	Total Deaths:	Date:	Total Cases:	Total Deaths:
March 18th	2556	17	April 2nd	92743	2326
March 19th	5699	34	April 3rd	103158	2955
March 20th	8530	56	April 4th	114173	3549
March 21st	12323	75	April 5th	123159	4151
March 22nd	16893	148	April 6th	131829	4758
March 23rd	21727	184	April 7th	140385	5489
March 24th	26389	269	April 8th	151078	6268
March 25th	33038	366	April 9th	161807	7067
March 26th	39141	461	April 10th	172972	7845
March 27th	46124	596	April 11th	181825	8626
March 28th	53520	834	April 12th	190288	9385
March 29th	59454	966	April 13th	196146	10056
March 30th	67387	1162	April 14th	203377	14612
March 31st	76051	1398	April 15th	214832	15648
April 1st	84046	1781	April 16th	227625	16130

Fig. 2: Total Case and Mortality Data for New York, from 18 March, 2020 through 16 April, 2020

For this model, we had to assume that the information provided was accurate–particularly concerning the total number of infections and thus, the total number of deaths from the virus. Furthermore, we assumed that each new case and new death was reported on onset date, and not delayed.

We used the online graphing tool, *Desmos* and *Microsoft Excel* to calculate the regression formula for this data. By using the same data for each tool, we were able to obtain the same regression formula using a least squares regression model. Furthermore, we chose to use a logistic model to calculate the regression formula:

$$TotalCases = \frac{A}{1+Be^{-CD}},$$

where A(carrying capacity), B and C(growth rate) are constants; D is the days since 18 March, where "1" corresponds to 18th March; and $TotalCases$ is the total

Florida COVID–19 Total Cases and Total Deaths Data

Date:	Total Cases:	Total Deaths:	Date:	Total Cases:	Total Deaths:
March 18th	328	9	April 2nd	9003	144
March 19th	434	10	April 3rd	10262	170
March 20th	565	12	April 4th	11539	195
March 21st	762	14	April 5th	12350	221
March 22nd	1005	16	April 6th	13629	253
March 23rd	1228	20	April 7th	14747	296
March 24th	1471	23	April 8th	15698	322
March 25th	1977	23	April 9th	16826	369
March 26th	2483	29	April 10th	17967	419
March 27th	3198	48	April 11th	18983	445
March 28th	4038	56	April 12th	19893	461
March 29th	4948	60	April 13th	21019	499
March 30th	5737	68	April 14th	21623	570
March 31st	6741	85	April 15th	22519	614
April 1st	7549	99	April 16th	23338	668

Fig. 3: Total Case and Mortality Data for Florida, from 18 March, 2020 through 16 April, 2020

Colorado COVID–19 Total Cases and Total Deaths Data

Date:	Total Cases:	Total Deaths:	Date:	Total Cases:	Total Deaths:
March 18th	218	49	April 2nd	3742	418
March 19th	220	2	April 3rd	4181	439
March 20th	341	121	April 4th	4565	384
March 21st	477	136	April 5th	4950	385
March 22nd	594	117	April 6th	5178	228
March 23rd	721	127	April 7th	5429	251
March 24th	909	188	April 8th	5655	226
March 25th	1085	176	April 9th	6202	547
March 26th	1460	375	April 10th	6511	309
March 27th	1759	299	April 11th	6892	381
March 28th	2025	266	April 12th	7307	415
March 29th	2251	226	April 13th	7693	386
March 30th	2593	342	April 14th	7948	255
March 31st	2965	372	April 15th	8280	332
April 1st	3324	359	April 16th	8675	395

Fig. 4: Total Case and Mortality Data for Colorado, from 18 March, 2020 through 16 April, 2020

number of positively infected cases.

Beginning with New York, we found $A = 257144$, $B = 43.764$, and $C = 0.1974$ with an average error percentage of 16.67% and a median error of 8.22% (see figure 5).

Using this same method on New York's total mortality rate from the virus, as depicted in figure 6.

Repeating this process on data from Florida and Colorado, we were able to model total case (figures 7 and 8) and mortality rates (figures 9 and 10) in the remaining two states that closely match the current trends, and that we feel accurately predict future trends in viral outbreak, provided there are no more surges in cases in these three regions.

We believe that this logistic model best represents current

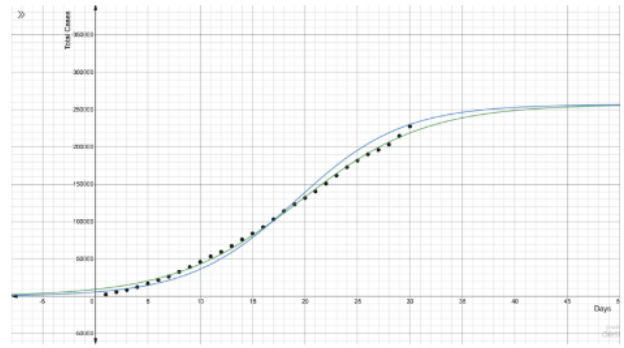

Fig. 5: Logistic Model of predicted total cases in New York using data from 18 March, 2020 through 16 April, 2020

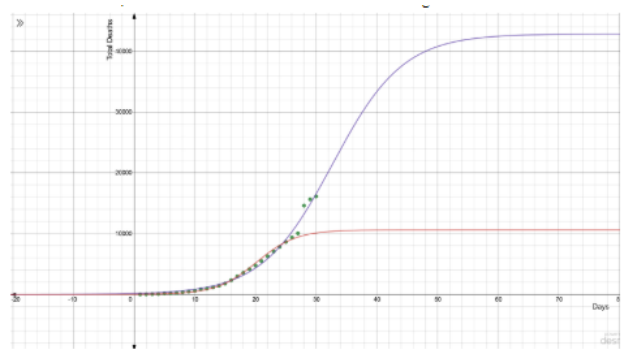

Fig. 6: Logistic Model of predicted total deaths in created for New York using data from 18 March, 2020 through 16 April, 2020

and future trends of the virus because it accurately describes current rates and shows promise for implementation in agent-based modeling simulations.

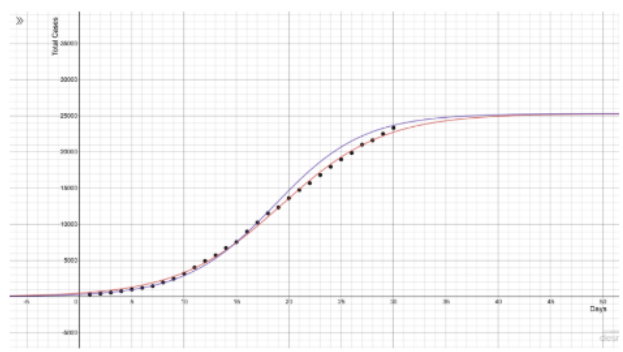

Fig. 7: Logistic Model of predicted total cases in Florida using data from 18 March, 2020 through 16 April, 2020

D. Simulation

Finally, we created a agent-based model, utilizing the provided data and the logistic regression model we derived, in order to simulation and visualize our findings. This models follows guidelines from our conceptual model, such as assumptions, parameters, and state variables; while also deriving information from the mathematical model (see figures 11, 12, and 13). It should be noted that this simulation focuses on the largest cities in each region

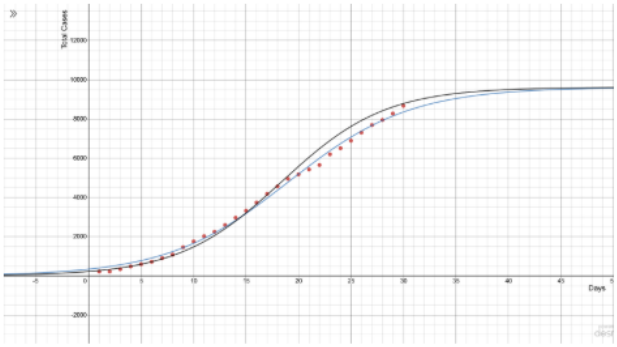

Fig. 8: Logistic Model of predicted total cases in Colorado using data from 18 March, 2020 through 16 April, 2020

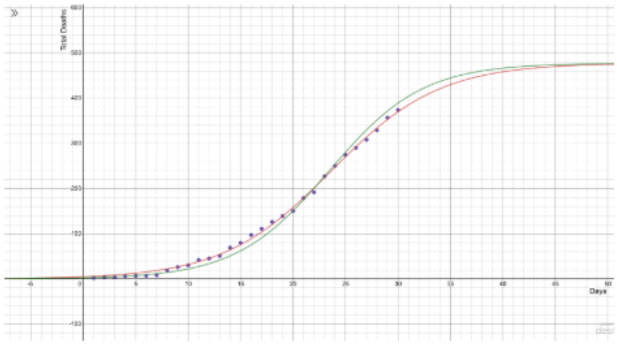

Fig. 9: Logistic Model of predicted total deaths in Colorado using data from 18 March, 2020 through 16 April, 2020

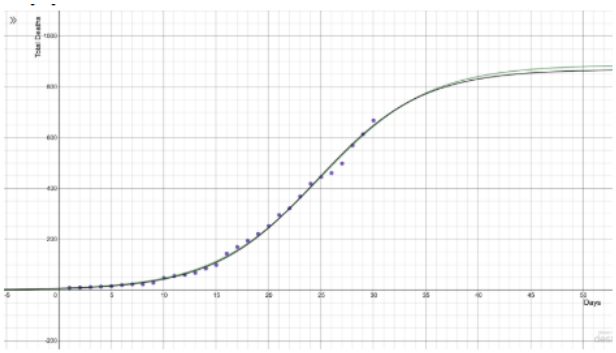

Fig. 10: Logistic Model of predicted total deaths in Florida using data from 18 March, 2020 through 16 April, 2020

(Denver, Colorado; Jacksonville, Florida; New York City, New York), rather than the state as a whole, and does not account for population density.

III. Results

A. Analysis

Using the model derived, we can predict the following for each state:

New York By our model, New York should not experience any new cases 122 days after our initial date, or 16 July, 2020, when positive cases total 257,144 (see figure 10). This is out of New York's 8.399 million total population, making the positive case totals on 16 July, 2020 3% of New York's Population. Additionally, cumulative death totals should stop

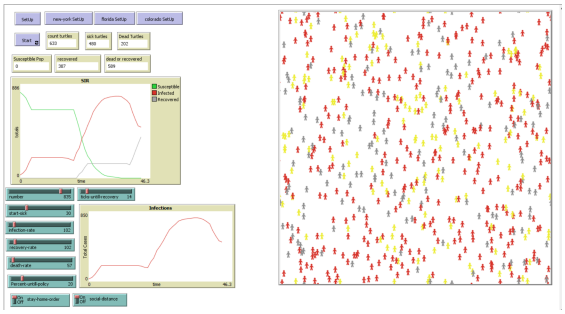

Fig. 11: Simulation of conceptual model in New York City, NY using data from 18 March, 2020 through 16 April, 2020

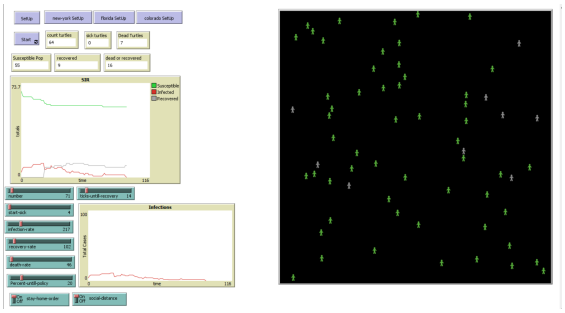

Fig. 12: Simulation of conceptual model in Jacksonville, FL using data from 18 March, 2020 through 16 April, 2020

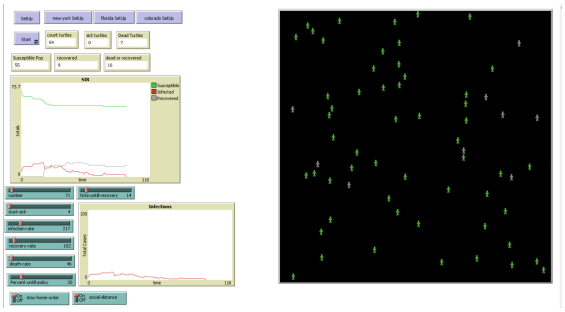

Fig. 13: Simulation of conceptual model in Denver, CO using data from 18 March, 2020 through 16 April, 2020

growing after they reach 10,616 individuals on 14 May, 2020, 57 days after our initial date. From the total population, this is 0.1%.

Florida By our model, Florida should see no new cases 68 days after the initial date, or 25 May, 2020, when case rates total 25,314 (see figure 12), or 0.1% of Florida's total population of 21.48 million. Furthermore, we anticipate that 66 days after our initial date, death totals will cease to rise. This means Florida will have a total of 869 deaths on 23 May, 2020, 0.004% of the state's total population.

Colorado According to this model, Colorado should not see any new cases 70 days after our initial formula date. Thus, we anticipate that the virus will cease spreading on 27 May, 2020, after infecting 0.1% of Colorado's population of 5.759 million; a total of 9,620 individuals (see figure 14). Similarly, we should see that there are no new deaths 56 days after our initial date, or 12 May, 2020, when cumulative deaths total 477 individuals, or 0.008% of the

total population.

When comparing these figures for each region, we see that New York will experience significantly higher infection rates than Florida and Colorado (2% more of the population testing positive for infection), Similarly, New York will experience increased mortality from the virus, at 1% of the population dying from the infection, where Colorado will lose 0.008% of the population, and Florida with 0.004%. These numbers may be explained by population density, as the highest population city in New York,New York City contains 26,403 people per square mile, while Jacksonville only has 1,110 people per square mile, and Denver has 4,044 people per square mile. Even though social-distancing regulations may be in place, it may be difficult for New York City residents to maintain the proper social-distancing in closer proximity to each other. It may also be that there are more individuals who must still go out in New York, or that individuals in New York are not following social-distancing measures, but we were unable to find data to support these hypotheses. It's difficult to determine if social-distancing measures have been effective in reducing total case and mortality rates in any of these states; however, it does appear that these measures have been effective in slowing the spread of the virus, and thus alleviating pressure on healthcare institutions.

These forecasts follow the assumptions and trends of our mathematical model; all of these forecasts assume that there are no surges in cases; however, officials anticipate that case and death rates may rise when social-distancing guidelines are relaxed. Furthermore, this analysis appears on par with other credible sources for each state.

IV. Discussion

A. Model Validation

To validate this model and simulation, we will first examine New York, and particularly New York City. New York City has the highest population of the regions which we evaluated, and has been practicing social-distancing and following stay-at-home orders since 22 March, 2020. We see a resemblance between the graph of actual coronavirus cases in New York City through the month of April and our simulation model. That is to say, at the beginning of the month of April, there is a gradual increase that fluctuates into a large spike and continues to spike and fluctuate while new case and mortality rates decrease. Our simulated model provides very similar data, though our spikes indicate less surges in cases.

Furthermore, when examining Jacksonville, Florida, we must first note that Florida has been mandated social-distancing as well as a stay-at-home order. We saw less similarities between actual case and mortality trends through the month of April and our predicted trends through the simulated model, an error that may be due to more randomization within the simulated model, including not accounting for population density or mass gathering locations.

When examining Denver, CO, it is immediately evident that our predicted simulated model does not correspond with the actual cases for the month of April. Colorado mandated stay-at-home orders and social-distancing, effective 26 March, 2020, which have been relaxed as of 26 April, 2020. We do not believe that the spike in actual cases on April 26, 2020 was due to the lessening of social-distancing regulations; however, it is possible that individuals have not complied with social-distancing regulations as the month progressed.

Invariants in our simulated model include recovery time, immune population (those who could no longer contribute to infection rates through recovery or death), and the rate of recovery. These figures are comparable to the actual data provided on the virus and based on actual data provided by credible sources.

V. Conclusions

A. Limitations

There were several limitations that caused us to make assumptions and may be areas of error within this study. All of our assumptions listed may mean that our predictions are not entirely accurate, particularly if cases surge after the easing of social-distancing protocols. Furthermore, our data only takes into account cases which have been tested and thus tested positive for the virus. There is a high likelihood that some individuals are asymptomatic carriers or experienced mild symptoms and thus did not receive a test. Additionally, testing in the United States has not been widely available, and therefore, even those who wished to seek testing may not have been able to do so. These circumstances may have led to under-reporting on case and mortality rates in each of our regions, and across the nation.

B. Implications and Further Research

There are many areas of future research that may be derived from this paper. We may simply continue to track and predict the virus, perhaps taking into account more assumptions, such as individuals who wear personal protective equipment (PPE); movement trends between population, particularly to spaces of mass gathering, such as grocery stores and churches; and virus behaviors including life-span on surfaces and transmission mode. Furthermore, we may wish to take into account for variables which may account for higher mortality rates across the country that are not caused directly by virus infection, but may be a result of economic recession or other societal factors that were affected by the virus.

In this model, we aimed to predict total case and mortality rates of the COVID-19 virus in several different communities across the United States, according to their current trends and the mitigation measures taken by each region. We have concluded that we are nearing the end of the outbreak, and that the social-distancing guidelines set in place by local governments may have had some effect on lower the cumulative number of cases and deaths, as well as slowing the rate at which these totals rise. From these conclusions, we encourage decision makers in government to continue implementing social-distancing

guidelines for the sake of healthcare institutions and the general population. However, there are several factors that we were not able to consider, such as how the economic downfall may effect the population, and the risk of postponing ”non-essential” surgeries and medical procedures, as well as the psychological toll of social distancing measures on the general population. It is encouraged that these variables are investigated further in order to better model the future of the outbreak and the overall effect on society in the United States.

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